

True bearings



1 State the true bearing of the following:

a Southwest

c West-Northwest

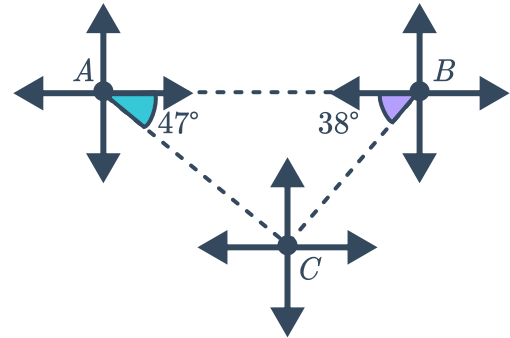
e Northeast

b Northwest

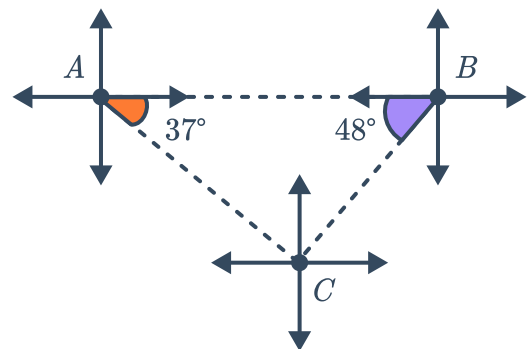
d South-Southeast

f Southeast

2 Determine the true bearing of point B from point A in the following bearings diagram:

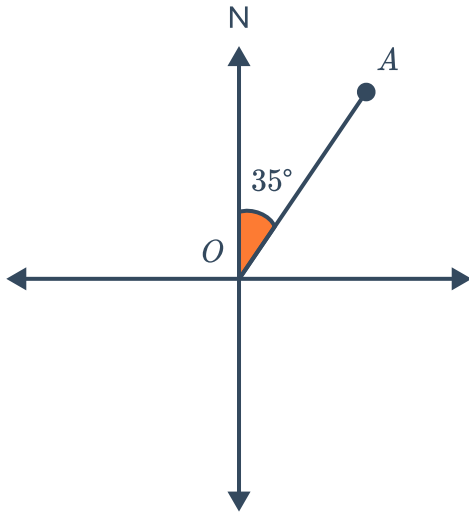


3 Determine the true bearing of point A from point B in the following bearings diagram:

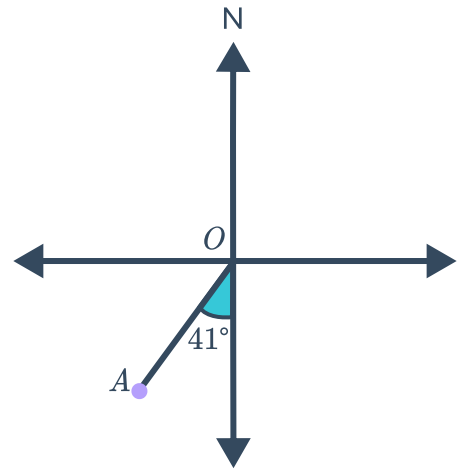


4 Determine the true bearing of point A from O in the following diagrams:

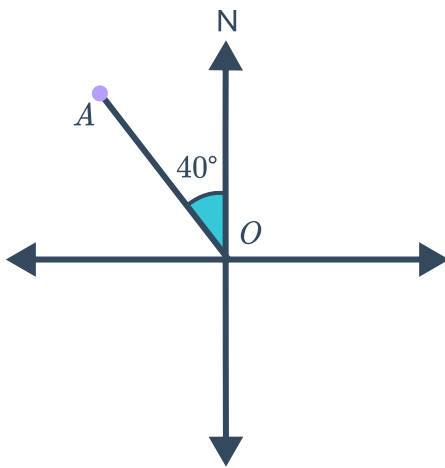
a



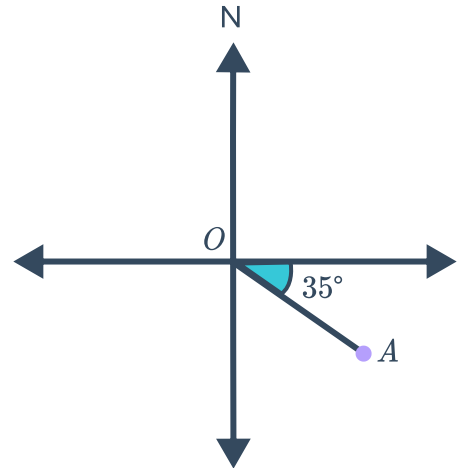
b



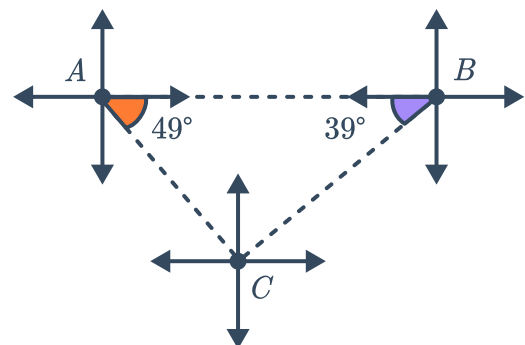
c



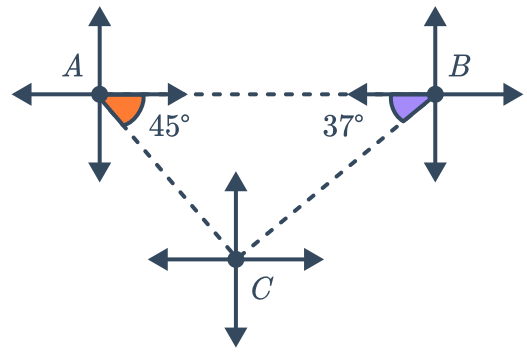
d



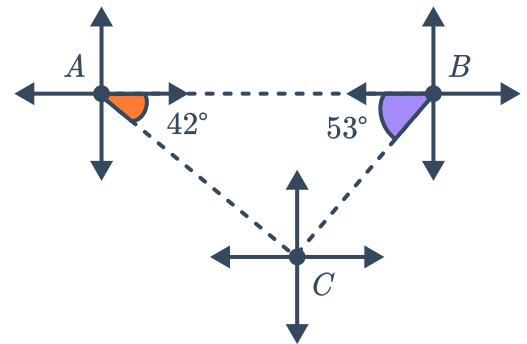
5 In the figure below, point B is due East of point A . Find the true bearing of point A from point C .



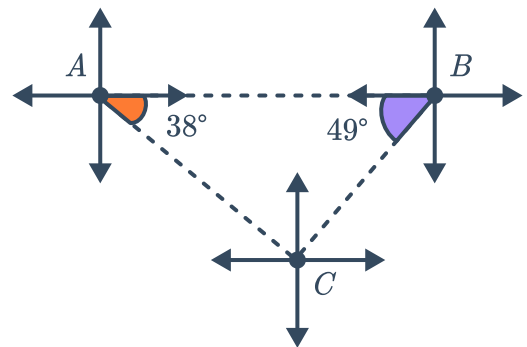
- 6 In the following diagram, find the true bearing of point B from point C .



- 7 In the following diagram, find the true bearing of point C from point A .

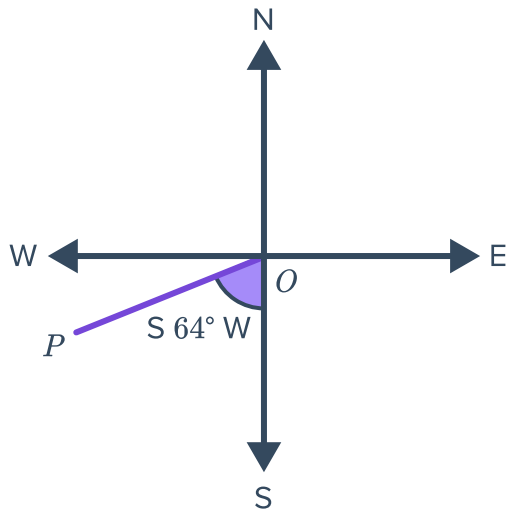


- 8 In the following diagram, find the true bearing of point C from point B .

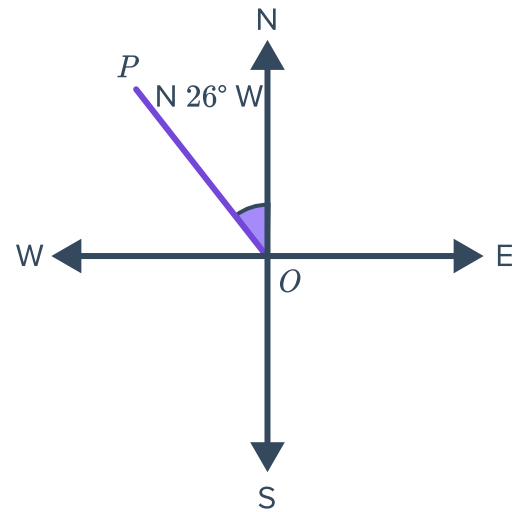


- 9 Determine whether the following diagrams show a true bearing of 296° T for the point P from the origin O .

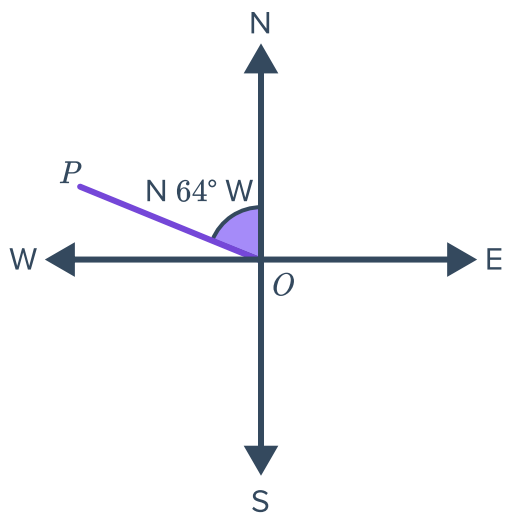
a



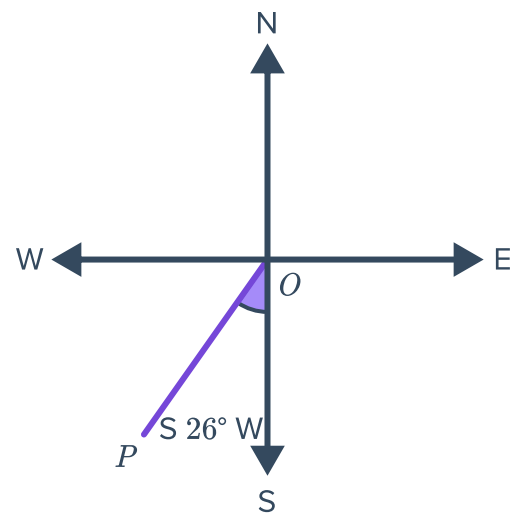
b



c

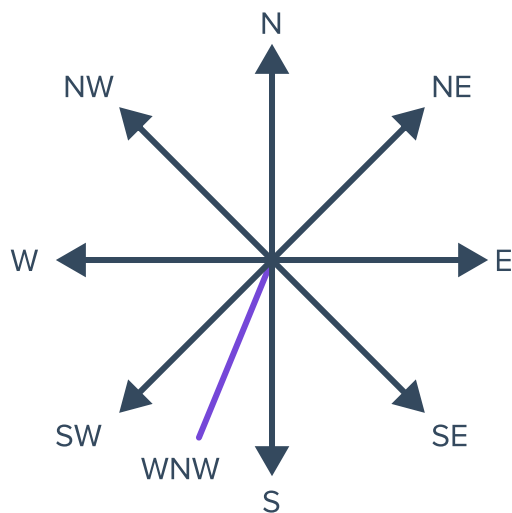


d

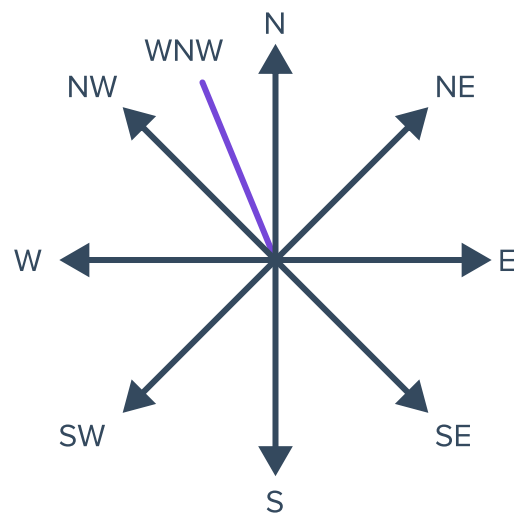


10 Determine whether the following diagram  lay *WNW*:

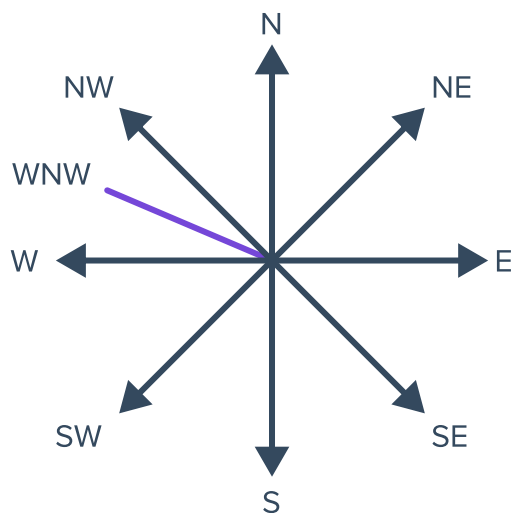
a



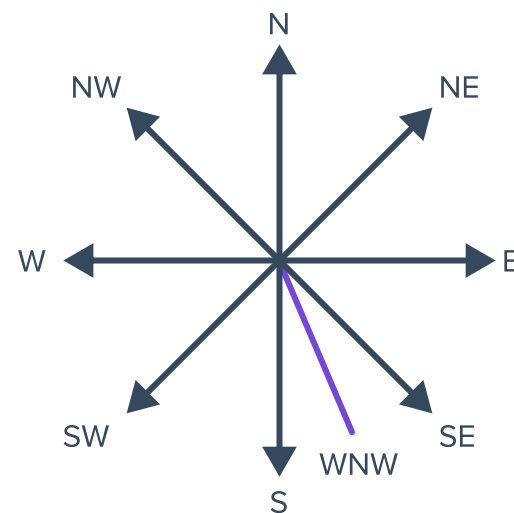
b



c



d



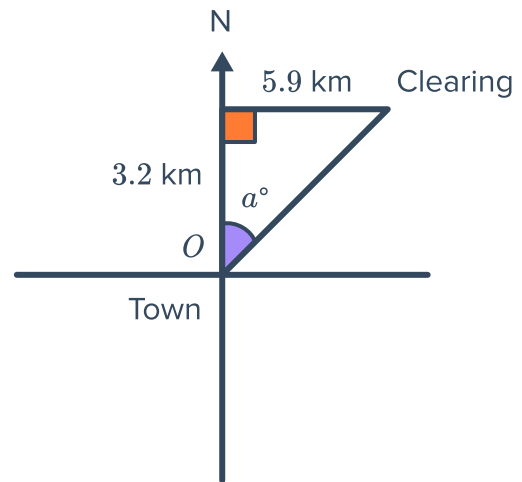
11 If a person walks at a compass bearing of *SW*, and then turns around to go back in the direction they came, what direction are they heading now?



Applications

12 Consider the following diagram:

If the bearing of the clearing from the town is a° , find a to the nearest degree.

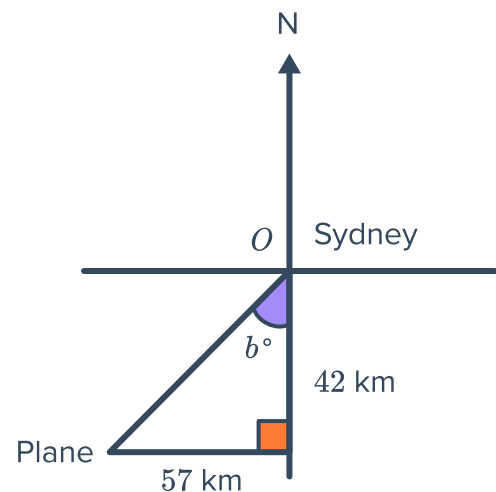


13 An aeroplane sets off on a bearing of 051°T . After some time, it needs to turn back and head to its original position. On what true bearing must it travel?

14 A boat sets off on a bearing of 125°T . After some time, it needs to turn back and head to its original position. On what true bearing must it travel?

15 Shortly after take-off, a plane is 42 km south and 57 km west of the airport in Sydney that it left from:

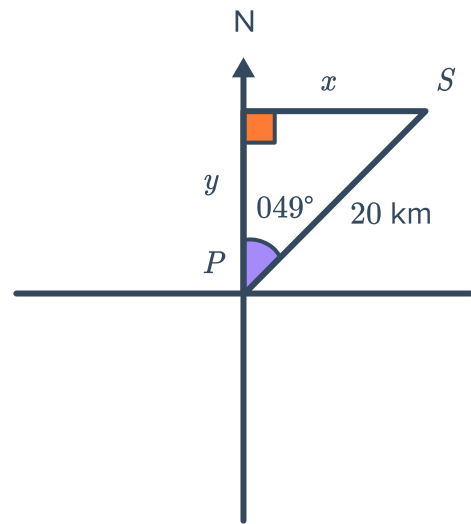
- a Find the size of the angle marked b , to one decimal place.
- b Find the true bearing of the plane from the airport.



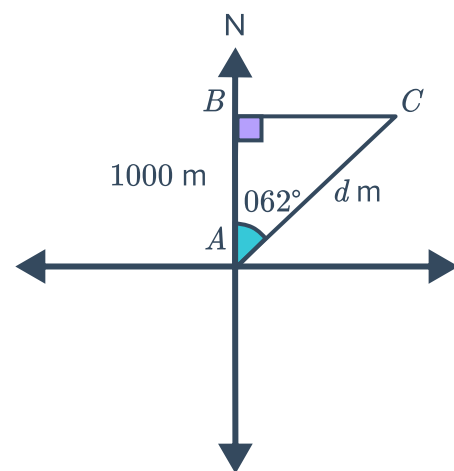
- 16 The position of a ship S is given to be 20 km from P , on a true bearing of 049° T. The position of the ship can also be given by its (x, y) coordinates.



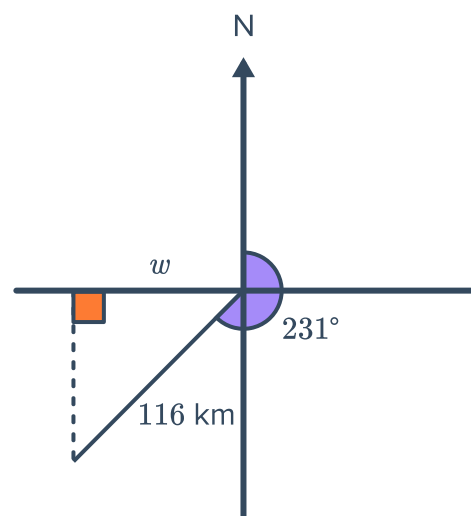
- Find the ship's x -coordinate to one decimal place.
- Find the ship's y -coordinate to one decimal place.



- 17 On an orienteering course, Beth runs 1000 metres north from point A to point B , then turns east and runs to point C . If the true bearing of C from A is 062° T and the direct distance Beth must run to get back to point A is d metres, find d to the nearest metre.



- 18 Luke sailed for 116 km on a bearing of 231° . If w is how many kilometres west he has sailed from his starting point, find w to one decimal place.

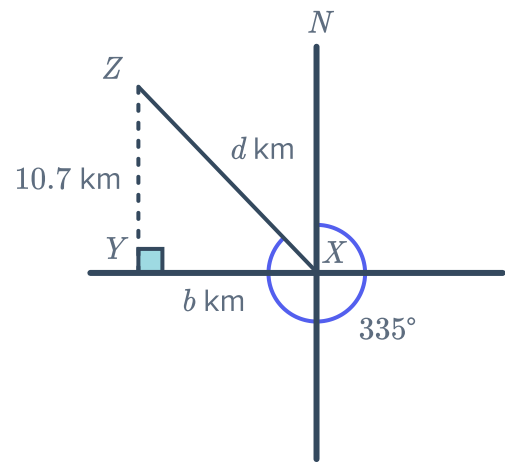


- 19 During a rescue search, a helicopter flew west from point X to point Y , then changed course and flew 10.7 km north to point Z .



If point Z is on a bearing of 335° T from point X :

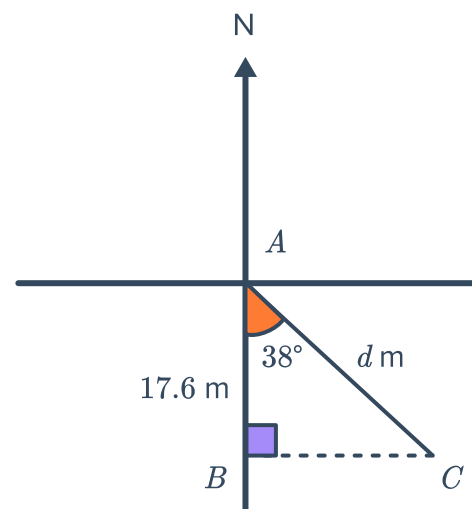
- Find the size of $\angle YXZ$.
- If the distance from point Y to point X is b km, find b to one decimal place.
- If the distance that the helicopter must fly between point Z and point X is d km, calculate d to one decimal place.



- 20 Three television presenters are practising their navigation skills before heading off on an expedition to a remote location.

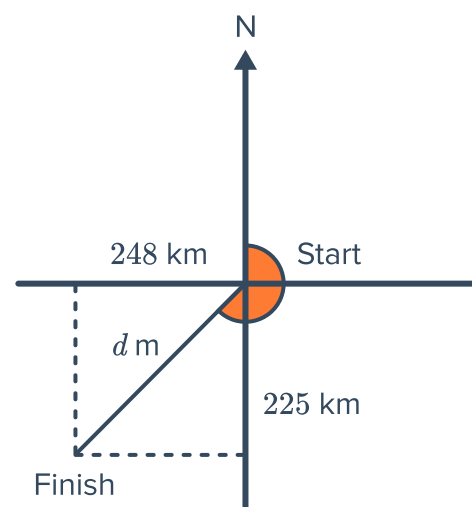
Belinda at point B is positioned 17.6 m south of Amelia at point A . Carl at point C is due east of Belinda and on a bearing of 142° from Amelia.

If Amelia and Carl are d m apart, find d to one decimal place.



- 21 A yacht sailed in a direction so that its final position was 248 km west and 225 km south of its starting point.

- If the true bearing on which the yacht sailed is b° , find the value of b to one decimal place.
- If the boat has sailed a total of d m, find the value of d to one decimal place.



22 In remote locations, photographers must keep track of their position from their base. One morning a photographer sets out from base, represented by point B , to the edge of an ice shelf at point S on a bearing of 055° . She then walked on a bearing of 145° to point P , which is 916 m due east of base.

- a** State the size of $\angle BSP$.
- b** Find the distance BS , to one decimal place.
- c** Find the distance SP , to one decimal place.
- d** If the photographer walks directly back to her base from point P , determine the total distance she would have travelled. Round your answer to one decimal place.

